

Day 20 - Lecture 20.19: Uniform Distribution

 Python 3.10+

PDF notes.pdf

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1. Overview & Principle of Maximum Entropy

The **Continuous Uniform Distribution** $\mathcal{U}(a, b)$ assigns perfectly equal probability density across all sub-intervals of equal length within the support $[a, b]$. Under bounded support with no prior information, the uniform distribution maximizes Shannon entropy.

Support Interval $[a, b]$ → Constant Density $f(x) = 1 / (b - a)$ → Total Area $(b - a) * (1 / (b - a)) = 1.0$

2. Mathematical Formalism

Probability Density Function (PDF):

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

Cumulative Distribution Function (CDF):

$$F(x) = \begin{cases} 0 & \text{for } x < a \\ \frac{x-a}{b-a} & \text{for } a \leq x \leq b \\ 1 & \text{for } x > b \end{cases}$$

Mathematical Derivation of Moments:

Expected Value (Mean μ):

$$\mathbb{E}[X] = \int_a^b x \cdot \frac{1}{b-a}, dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

Second Moment $\mathbb{E}[X^2]$:

$$\mathbb{E}[X^2] = \int_a^b x^2 \cdot \frac{1}{b-a}, dx = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

Variance (σ^2):

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{a^2 + ab + b^2}{3} - \frac{a^2 + 2ab + b^2}{4}$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} \quad \text{and} \quad \sigma = \frac{b-a}{\sqrt{12}}$$

3. Python Implementation & Simulation

```
import numpy as np
import matplotlib.pyplot as plt

# Generate 1,000,000 samples from U(0, 10)
a, b = 0.0, 10.0
n_samples = 1_000_000
values = np.random.uniform(a, b, n_samples)

mean_ana = (a + b) / 2
var_ana = ((b - a) ** 2) / 12

print(f"Uniform U({a}, {b}) Parameters:")
print(f"  Analytical Mean:      {mean_ana:.4f} | Empirical: {np.mean(values):.4f}")
print(f"  Analytical Variance: {var_ana:.4f} | Empirical: {np.var(values):.4f}")
print(f"  Theoretical PDF:      1/(b-a) = {1/(b-a):.4f}")
```

4. Key Takeaways & Interview Points

- **The "Over 12" Divisor:** The denominator 12 in $\text{Var}(X) = \frac{(b-a)^2}{12}$ is a favorite interview derivation test.
- **Neural Network Weight Initialization:** He and Xavier Uniform initializations sample weights from $\mathcal{U}\left(-\sqrt{\frac{6}{n_{\text{in}}+n_{\text{out}}}}, +\sqrt{\frac{6}{n_{\text{in}}+n_{\text{out}}}}\right)$ to maintain variance across layers.